

1. Build and evaluate a **Random Forest Classifier** to predict whether a tumor is malignant or benign. Use the Breast Cancer dataset from `sklearn`.

Evaluate the model using:

- Accuracy
- Confusion Matrix
- Precision, Recall, F1-score
- ROC-AUC score

2. Build a Logistic Regression model to predict whether a patient has diabetes (1) or not (0).

1. Load the dataset into a pandas DataFrame.
2. Assign appropriate column names.
3. Check for missing or zero values.
4. Split the dataset into train (80%) and test (20%).
5. Apply feature scaling.
6. Train a Logistic Regression model.
7. Evaluate using:
 - Accuracy
 - Confusion Matrix
 - Precision, Recall, F1-score
 - ROC-AUC score
8. Interpret model coefficients.

3. Build an **XGBoost classifier** to predict whether a passenger survived the Titanic disaster.

1. Load the dataset into a pandas DataFrame.
2. Assign appropriate column names.
3. Check for missing or zero values.
4. Split the dataset into train (80%) and test (20%).
5. Apply feature scaling.
6. Train a XGBoost model and include parameters
7. Evaluate using:
 - Accuracy
 - Confusion Matrix
 - Precision, Recall, F1-score
 - ROC-AUC score
8. Interpret model coefficients.

4. Build a **Decision Tree Classifier** to predict whether a patient has diabetes (0 = No, 1 = Yes). Use the **Pima Indians Diabetes Dataset**.

1. Load the dataset into a pandas DataFrame.
2. Assign appropriate column names.
3. Check for missing or unrealistic zero values and handle them properly.
4. Split the dataset into:

- 80% training set
- 20% testing set
- Use `random_state=42`

5. Define features (X) and target variable (y).

6. Train a `DecisionTreeClassifier` model.

7. Evaluate the model using:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-score

8. Train another Decision Tree with restricted depth (`max_depth=3`).

Compare the performance of both models.

Extract and display feature importance.